CHAPTER 2

ELECTROCHEMISTRY

CHAPTER ESSENCE AT A GLANCE:
Redox transformations producing electrical energy in galvanic cells, standard electrode potentials via Nernst equation, conductivity, and electrolysis.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Galvanic Cells & Standard Potentials:** Spontaneous chemical redox energy converts into electrical energy at separate electrodes, where oxidation occurs at the negative anode and reduction occurs at the positive cathode, generating a cell potential determined by standard electrode reduction values.

• **Nernst Equation & Thermodynamic Equilibrium:** Relates electrode and cell potentials to reaction quotient and ion concentrations at any temperature, directly linking cell electromotive force to standard Gibbs free energy and the equilibrium constant.

• **Electrolytic Conductivity & Concentration Trends:** Specific conductivity decreases upon dilution as fewer charge carriers occupy each unit volume; conversely, molar conductivity increases with dilution due to reduced ionic interferences in strong electrolytes and enhanced degree of dissociation in weak electrolytes.

• **Kohlrausch's Independent Migration Law:** At infinite dilution, each ion contributes a distinct and independent limiting molar conductivity to the electrolyte regardless of the co-ion, enabling exact determination of limiting molar conductivity and ionization constants for weak electrolytes.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n}\log\left(\frac{[\text{Products}]}{[\text{Reactants}]}\right)$$

Nernst Equation for galvanic cell potential at 298 K

$$\Delta G^{\circ} = -nFE^{\circ}_{\text{cell}} \quad \text{and} \quad \log K_c = \frac{nE^{\circ}_{\text{cell}}}{0.0591}$$

Standard Gibbs free energy change and equilibrium constant relations

$$\kappa = \frac{1}{R}\left(\frac{l}{A}\right) = G \cdot G^{*} \quad \text{and} \quad \Lambda_m = \frac{1000\kappa}{C}$$

Conductivity , cell constant G, and molar conductivity m*

$$\Lambda^{\circ}_m = \nu_+ \lambda^{\circ}_+ + \nu_- \lambda^{\circ}_- \quad \text{and} \quad \alpha = \frac{\Lambda_m}{\Lambda^{\circ}_m}$$

Kohlrausch's Law of independent ionic migration and degree of dissociation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse and m on dilution! decreases while m increases.

• **Alert:** [Valuation Rubric] Step 1 [Cell Reaction & n]: Write overall cell reaction and identify number of transferred electrons n .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Represent the Daniell cell and write the cell reactions occurring at anode and cathode. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Cell Representation]** Zn(s) | Zn^{2+}(aq) || Cu^{2+}(aq) | Cu(s). [0.5 Mark]

• **[Step 2: Reactions]** Anode (Oxidation): Zn(s) --> Zn^{2+}(aq) + $2e^-$. Cathode (Reduction): Cu^{2+}(aq) + $2e^-$ --> Cu(s). Overall: Zn(s) + Cu^{2+}(aq) --> Zn^{2+}(aq) + Cu(s) (n = 2). [1.5 Marks]

• **Key Exam Result:** Zn + Cu^{2+} --> Zn^{2+} + Cu (E° = +1.10 V)

2. State the Nernst equation for a general electrode reaction and a general galvanic cell. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: General Electrode]** M^{n+} + ne^- --> M(s): E = E° - (2.303 RT divided by nF) $\log_{10}$(1 divided by $[M^{n+}]$). At 298 K: E = E° - (0.0591 divided by n) $\log_{10}$(1 divided by $[M^{n+}]$). [1.5 Marks]

• **[Step 2: Galvanic Cell]** aA + bB --> cC + dD: E cell = E° cell - (0.0591 divided by n) $\log_{10}([C]^c [D]^d$ divided by $[A]^a [B]^b$). [1.5 Marks]

• **Key Exam Result:** E cell = E° cell - (0.0591 divided by n) log Q

3. Relate standard cell potential E° {cell} with equilibrium constant K c and Gibbs free energy \ΔG^°. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Gibbs Free Energy]** ΔG° = -n F E° cell. [1.5 Marks]

• **[Step 2: Equilibrium Constant]** E° cell = (2.303 RT divided by nF) $\log_{10}$ K c = (0.0591 divided by n) $\log_{10}$ K c at 298 K implies $\log_{10}$ K c = n E° cell divided by 0.0591. [1.5 Marks]

• **Key Exam Result:** ΔG° = -n F E° cell | log K c = n E° cell divided by 0.0591

4. Define resistivity, conductivity (specific conductance), and molar conductivity with SI units. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Resistivity & Conductivity]** Resistivity ρ = R · A divided by l (Ω m). Conductivity = 1 divided by ρ = (1 divided by R) · (l divided by A) = G · G* (S m^{-1} or Ω^{-1} cm^{-1}). [1.5 Marks]

• **[Step 2: Molar Conductivity]** Conducting power of all ions from 1 mole electrolyte: $_m$ = divided by C = (1000 ×) divided by M (S cm^2 mol^{-1}). [1.5 Marks]

• **Key Exam Result:** = G · G* (S m^{-1}) | $_m$ = (1000) divided by M (S cm^2 mol^{-1})

5. Explain the variation of conductivity and molar conductivity with dilution for strong and weak electrolytes. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Conductivity with Dilution]** Conductivity decreases with dilution for both strong and weak electrolytes because number of current-carrying ions per unit volume decreases. [1.5 Marks]

• **[Step 2: Molar Conductivity $_m$]** $_m$ increases with dilution. Strong electrolytes: increases slowly due to decrease in interionic attractions (Debye-Huckel-Onsager equation). Weak electrolytes: increases steeply at high dilution due to sharp increase in degree of dissociation α (Kohlrausch curve). [1.5 Marks]

• **Key Exam Result:** decreases; $_m$ increases upon dilution

6. State Faraday's first and second laws of electrolysis. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: First Law]** Mass of substance deposited or liberated is directly proportional to quantity of electricity passed: m = Z Q = Z I t. [1 Mark]

• **[Step 2: Second Law]** When same current passes through different electrolytes in series, masses deposited are proportional to their chemical equivalents: m_1 divided by m_2 = E_1 divided by E_2. [1 Mark]

• **Key Exam Result:** m = Z I t | m_1 divided by m_2 = E_1 divided by E_2

7. Describe the construction, chemical reactions, and working of Lead Storage Battery during discharge and recharge. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Discharge Reactions]** Anode: $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$. Cathode: $PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$. Overall: $Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$. [2 Marks]

• **[Step 2: Recharge Reactions]** All reactions reverse; density of H_2SO_4 restores to 1.30 g/mL: $2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$. [2 Marks]

• **Key Exam Result:** $Pb + PbO_2 + 2H_2SO_4 <= => 2PbSO_4 + 2H_2O$